



WEB DEVELOPMENT - HTML Fundamentals

Focus: Structure of HTML document
Text formatting, links, images
Lists and tables





What You'll Learn Today

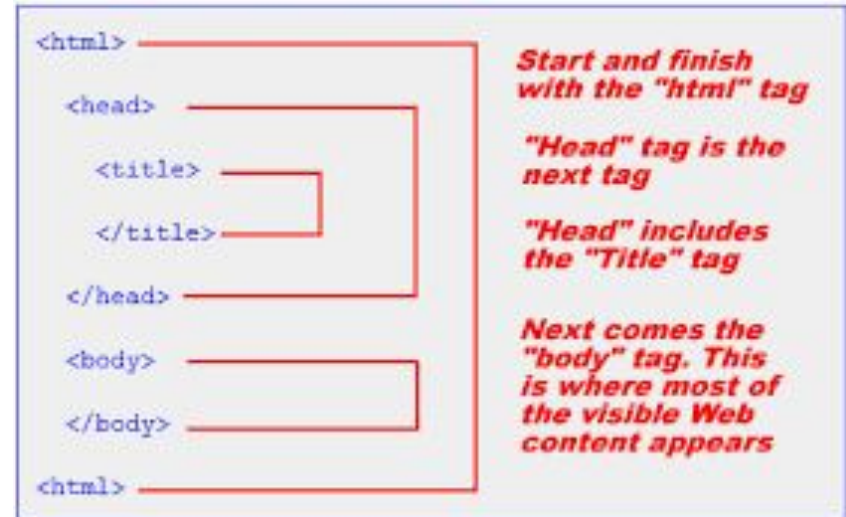
By the end of this lesson, students should be able to:

1. Describe the basic structure of an HTML document.
2. Apply text formatting tags to create readable content.
3. Create hyperlinks and embed images in web pages.
4. Construct ordered and unordered lists.
5. Design basic tables to display structured data.



Structure of an HTML Document

An HTML document provides the foundational structure for web pages. Understanding its anatomy is essential for creating well-formed, functional, and accessible websites. Each HTML document is organized into specific elements, each with a distinct purpose.





Structure of an HTML Document

`<!DOCTYPE html>` – Document Type Declaration

This declaration appears at the very beginning of an HTML file and informs the web browser about the version of HTML being used. For modern web development, `<!DOCTYPE html>` specifies HTML5, ensuring that browsers interpret the content correctly and consistently.

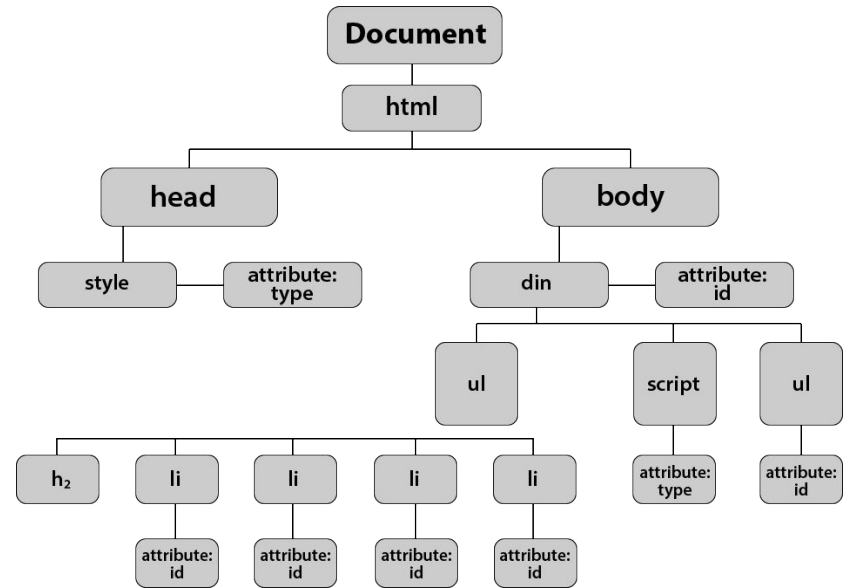




Structure of an HTML Document

`<html>` - Root Element

The `<html>` tag serves as the root container for the entire document. It wraps all other elements, including the `<head>` and `<body>`. Attributes such as `lang` can be added to specify the language of the document, improving accessibility and search engine optimization.





Structure of an HTML Document

`<head>` - Metadata and Resources

The `<head>` section contains metadata and resources that are not displayed directly to users but are crucial for browser rendering and site functionality. Common elements within the head include:

- `<title>`: Defines the page title displayed in browser tabs.
- `<meta>`: Provides metadata such as character encoding and viewport settings.
- `<link>`: Links external stylesheets or icons.
- `<script>`: References external JavaScript files.

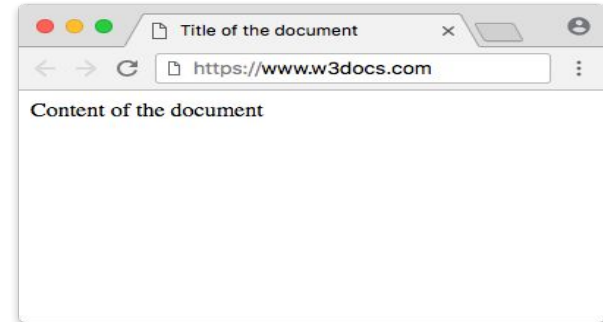
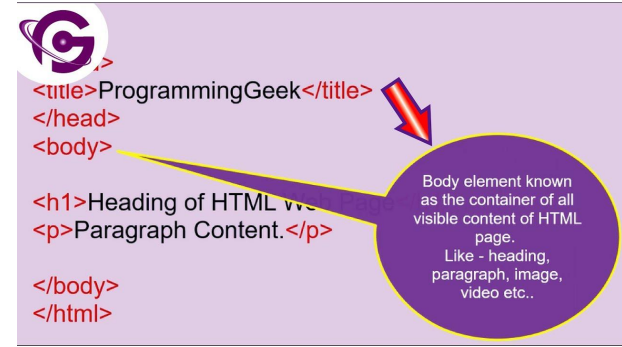
```
6 <head>
7   <title> html page </title>
8   <meta charset="utf-8">
9   <meta name="viewport" content="width=device-width, initial-scale=1">
10  <meta name="author" content="NTC">
11  <meta name="robots" content="index, follow">
12  <link rel="contents" href="#top">
13 </head>
14 <body>
```




Structure of an HTML Document

`<body>` - Visible Content

The `<body>` section contains all content that users see and interact with on the web page. This includes text, images, links, tables, forms, multimedia elements, and interactive components. Proper organization and semantic markup in the body improve usability, accessibility, and search engine optimization.





Text Formatting, Links, and Images

Creating visually organized and interactive content is a fundamental part of web development. HTML provides specific elements to format text, create hyperlinks, and display images effectively.

Formatting Text and Adding Links





Text Formatting, Links, and Images

Text Formatting

HTML offers a variety of **text formatting tags** to structure and emphasize content:

- **Headings (<h1>—<h6>):** Define headings in decreasing order of importance. <h1> is the main title, while <h6> is the least prominent heading.
 - **Paragraphs (<p>):** Used for blocks of text, providing proper spacing and readability.
 - **Emphasis (and):** indicates strong importance (often rendered in bold), while denotes emphasis (typically italicized).
 - **Span ():** A generic inline container for styling or scripting a specific portion of text without affecting the surrounding content.
- These tags help structure content semantically, improve accessibility, and enhance visual presentation.

HTML TEXT FORMATTING

Tag	Purpose
 and 	To Bold Text
<i> and 	To italic Text
<sub>	Defines subscripted text
<sup>	Defines superscripted text
<small>	Defines smaller text
<big>	Defines bigger text
<ins>	Defines inserted text
	Defines deleted text
<mark>	Defines marked/highlighted text



Text Formatting, Links, and Images

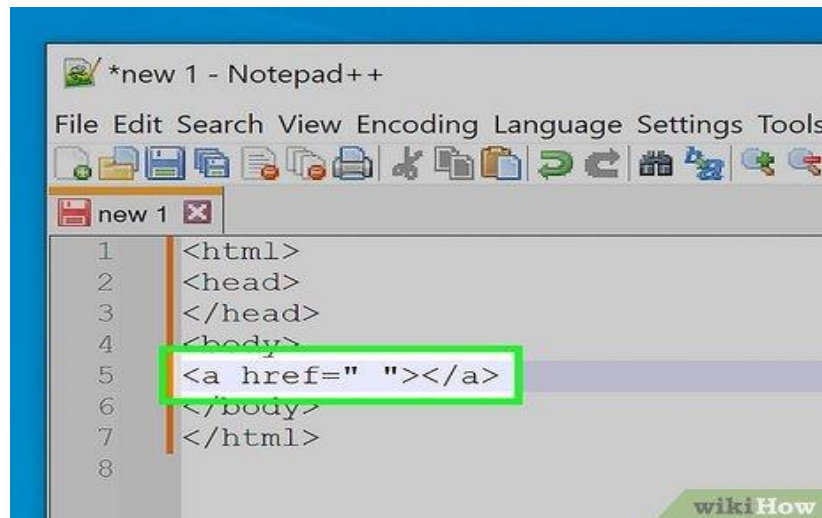
Hyperlinks

Hyperlinks allow users to navigate between pages or resources. The `<a>` tag is used to define a link:

- The `href` attribute specifies the destination URL.
- The text between the opening and closing `<a>` tags becomes clickable.
- Links can point to external websites, internal pages, email addresses, or downloadable files.

Hyperlinks are essential for creating interconnected content and facilitating easy navigation on the web.

```
<a href="https://example.com">Visit Example</a>
```





Text Formatting, Links, and Images

Images

Images are added using the `<img>` tag:

- **src**: Specifies the path to the image file.
- **alt**: Provides alternative text that describes the image for accessibility and search engines.

Images enhance visual appeal, illustrate concepts, and improve user engagement. Proper use of the `alt` attribute also ensures compliance with accessibility standards.

```

```





Lists and Tables

HTML provides elements for organizing content into **lists** and **tables**, which improve readability, structure, and clarity on web pages.

Heading 1	Heading 2	Heading 3
1. Data	■ Data	A. Data
2. Data	■ Data	B. Data
3. Data	■ Data	C. Data

```
<ol type='A' >  
  <li> Data </li>  
  <li> Data </li>  
  <li> Data </li>  
</ol>
```



Lists and Tables

Lists

Lists are used to present items in a structured manner. HTML supports two main types of lists:

- **Ordered List ():** Displays items in a numbered sequence, useful when order matters, such as steps in a process.
- **Unordered List ():** Displays items with bullet points, suitable for general grouping of items without priority.
- **List Items ():** Each item in a list, whether ordered or unordered, is wrapped in an tag.

```
<ol>
  <li>First step</li>
  <li>Second step</li>
</ol>

<ul>
  <li>Apples</li>
  <li>Oranges</li>
</ul>
```

HTML Demo: RESET

HTML	CSS	OUTPUT
<pre>2 Item 1 3 Item 2 4 5 Subitem 2.1 6 Subitem 2.2 7 8 9 Item 3 10 11 Subitem 3.1 12 Subitem 3.2 13 14 15 16</pre>		• Item 1 • Item 2 ◦ Subitem 2.1 ◦ Subitem 2.2 • Item 3 1. Subitem 3.1 2. Subitem 3.2

Lists make content easy to scan, improving user experience and comprehension.



Lists and Tables

Tables

Tables are used to display **tabular data** in rows and columns, providing a clear way to organize related information. Key HTML table elements include:

- **<table>**: The main container for all table content.
- **<tr> (table row)**: Defines each row within the table.
- **<th> (table header)**: Defines header cells, usually displayed in bold and centered.
- **<td> (table data)**: Defines standard data cells in the table.

TABLES SYNTAX

```
<table>
  <tr>
    <td>cell 1</td>
    <td>cell 2</td>
    <td>cell 3</td>
  </tr>

  <tr>
    <td>cell 4</td>
    <td>cell 5</td>
    <td>cell 6</td>
  </tr>
</table>
```

HTML TABLES

cell 1	cell 2	cell 3
cell 4	cell 5	cell 6



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</tr>

<tr>
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  <td>cell 5</td>
  <td>cell 6</td>
</tr>
</table>
```

HTML TABLES

cell 1	cell 2	cell 3
cell 4	cell 5	cell 6

Tables are particularly useful for presenting structured data such as schedules, price lists, or statistical information, making it easier for users to compare values.



Summary



HTML forms the **foundation of web page structure** and enables the creation of organized, accessible, and interactive content. Key concepts include:

- Document Structure:** The `<!DOCTYPE html>`, `<html>`, `<head>`, and `<body>` elements provide the framework for all content.
- Text Formatting:** Headings, paragraphs, emphasis, and inline elements organize and highlight textual content.
- Hyperlinks:** The `<a>` tag allows navigation between pages, sites, and resources.
- Images:** The `<img>` tag displays visual content, with `alt` text supporting accessibility.
- Lists:** Ordered (`<ol>`) and unordered (`<ul>`) lists structure content for easy scanning.
- Tables:** `<table>`, `<tr>`, `<th>`, and `<td>` organize data in rows and columns for clear presentation.

Mastering these HTML elements provides a solid foundation for building **well-structured, readable, and user-friendly web pages**, forming the basis for more advanced web development concepts.



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